

PKG-SAT-001

Low-Cost Satellite Internet Service for Remote Regions — Viability & First Build

Engineering Concept Package — produced per MASTER_PROTOCOL.md

PACKAGE ID

PKG-SAT-001

PACKAGE MATURITY

DISCOVERY

DATE

2026-08-03

GOVERNANCE

MASTER_PROTOCOL.md (11 sections)

SCANNER

Law 27/28/29 compliant

REJECTED

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Typed status of this package

Package ID: PKG-SAT-001 **Package maturity:** DISCOVERY (per Law 1 — hypothesis stage, no evidence beyond the claim itself) **Date:** 2026-08-03 **Status:** **REJECTED** (the honest verdict is: do not build this yet; the next expensive risk is orbital, not technical)

This package was produced by following MASTER_PROTOCOL.md. The coder read MASTER_PROTOCOL.md and FAILURES.md, received the INPUT, and filled the 12 sections + Next Money Page. The protocol decided; the coder executed.

The INPUT is a business question ("is this viable?"), not a hardware design request. The package answers by identifying the next expensive risk and what the first \$25k should buy.

0. PURPOSE

What are we evaluating? A low-cost satellite internet service for remote regions, operated by a small company with limited capital. The question is not "can we build a satellite" — it is "should we spend the next dollar on this, and if so, on what?"

Primary objective: determine whether the idea is technically and economically viable for a small company with limited capital, and identify what to build first.

Success metric: "A customer (the founding team) can spend the next \$25k intelligently to eliminate the single highest-risk unknown." Status: **BLOCKED** — the highest-risk unknown is regulatory + launch-cost, not engineering. No amount of engineering spend resolves it.

Package maturity: DISCOVERY. No satellite has been designed. No frequency license has been applied for. No launch contract exists. The idea is a hypothesis.

1. REQUIREMENTS

ID	REQUIREMENT	CLASS	STATUS
R-001	Service provides ≥ 10 Mbps to remote users	MANDATORY	BLOCKED (requires satellite capability analysis)
R-002	User terminal cost $< \$500$	MANDATORY	BLOCKED (requires terminal design + RFQ)
R-003	Monthly service cost $< \$50/\text{user}$	DESIRABLE	BLOCKED (requires ground-segment cost model)
R-004	Coverage of at least 1 region (e.g., sub-Saharan Africa, rural South America)	MANDATORY	BLOCKED (requires orbital-mechanics analysis)
R-005	Launch within 24 months	ASPIRATIONAL	BLOCKED (launch market analysis required)
R-006	Capital requirement $< \$10\text{M}$ total	MANDATORY	BLOCKED (this is the kill question — see §11)
R-007	ITU frequency coordination completed	MANDATORY	BLOCKED (12-18 month process, \$200K-\$500K legal)
R-008	Ground station network (≥ 3 stations)	MANDATORY	BLOCKED (requires site survey + RFQ)

Conflicts: R-006 (capital $< \$10\text{M}$) is in direct conflict with the cost of a dedicated satellite launch (\$30M-\$60M for a small GEO; \$5M-\$15M for a rideshare LEO slot). A small company with limited capital cannot afford a dedicated launch. The only viable path is rideshare (SpaceX Transporter) or hosted payload — but this constrains the orbit and the schedule.

2. EVIDENCE

Existing services

SERVICE	ARCHITECTURE	USERS	TERMINAL COST	LESSON
Starlink (SpaceX)	LEO constellation, 5,500+ sats	2M+	\$599	Vertical integration from rocket to user; \$5B+ spent; not replicable by a small company
HughesNet (GEO)	GEO satellite, shared bandwidth	1M+	\$300	GEO has high latency (600ms); bandwidth shared; low per-user cost but poor experience
Iridium (LEO)	LEO constellation, 66 sats	200K+	\$1,300	Narrowband (voice + low-data); \$5B to build; bankruptcy in 1999, restructured
OneWeb (LEO)	LEO constellation, 648 sats	100K+	\$1,200	\$3B spent before bankruptcy in 2020; acquired by UK government + Bharti; now operational
AST SpaceMobile	LEO, direct-to-cell	Testing	\$0 (uses existing phones)	First direct-to-cell satellite call in 2023; \$400M+ raised; still pre-revenue

Failed services

FAILURE	CAUSE	LESSON
Iridium bankruptcy (1999)	\$5B debt, 50K users (projected 500K), phone cost \$3,000	The market was not ready; the capital model was wrong
OneWeb bankruptcy (2020)	\$3B spent, COVID market shock, no revenue	Constellation economics are brutal; you cannot stop spending once you start launching
LeoSat (cancelled 2019)	Could not raise \$2.5B; no anchor customer	Without an anchor customer, the capital does not close
Skybridge (cancelled 2000s)	Regulatory delays, cost overruns	ITU frequency coordination is a 12-18 month process that can kill the timeline

Patents

PATENT	SUBJECT	RELEVANCE
US 11,107,153 (SpaceX)	Phased-array user terminal	Starlink's terminal cost reduction; patent may restrict phased-array terminal designs
US 10,441,123 (Iridium)	LEO satellite handoff protocol	Relevant if designing a constellation; Iridium's handoff is proven
US 9,978,295 (OneWeb)	LEO satellite gateway architecture	Gateway architecture for LEO; may have claim families that restrict ground-segment design

Academic literature

SOURCE	FINDING
Crisp et al. 2022 (Acta Astronautica)	LEO constellation breakeven requires >100K subscribers at >\$50/month
del Portillo et al. 2019 (MIT)	GEO vs LEO capacity comparison: LEO has 10x lower latency but 3x higher capex per Gbps
ITU-R S.1525	LEO satellite coordination methodology — the regulatory baseline
FAA Launch Market Report 2024	Rideshare LEO launch cost: \$5,000/kg (SpaceX Transporter); dedicated small-lift: \$15,000/kg

Standards

STANDARD	SCOPE
ITU-R S.1525	LEO satellite coordination methodology
ITU Radio Regulations Article 22	Equivalent power flux-density limits (EPFD) for non-GSO systems
FCC Part 25	Satellite communications (US ground station licensing)
ETSI EN 302 307	DVB-S2 standard (satellite broadband)

Supplier data

COMPONENT	SUPPLIER	RANK	INDICATIVE COST
Satellite bus (smallsat, 100-200 kg)	Blue Canyon (US), Spaceflight (US), SSTL (UK)	E (manufacturer spec)	\$5M-\$15M per satellite
Launch (rideshare LEO, 200 kg)	SpaceX Transporter	E (public pricing)	\$1.1M (\$5,500/kg)
Launch (dedicated small-lift)	Rocket Lab Electron, Astra	E	\$7M-\$15M per launch
Ka-band transponder	Viasat (US), Airbus (EU)	E	\$2M-\$5M per satellite
Phased-array user terminal	SpaceX (not for sale), Kymeta (US)	E	\$500-\$1,500 per terminal

3. DECOMPOSITION

Subsystems

1. **Space segment** — satellite(s) in LEO or GEO; provides RF coverage
2. **Launch segment** — rocket + integration; delivers satellite to orbit
3. **Ground segment** — gateway stations + NOC (network operations center)
4. **User terminal** — antenna + modem + router at the customer site
5. **Regulatory** — ITU frequency coordination + national licenses + EPFD compliance
6. **Business operations** — billing, customer support, network management

Component cost breakdown (order-of-magnitude, LEO single-satellite)

COMPONENT	INDICATIVE COST	BASIS	STATUS
Satellite bus (100 kg)	\$5M-\$10M	CATALOG (Blue Canyon, SSTL)	ESTIMATED
Communication payload (Ka-band)	\$2M-\$5M	CATALOG (Viasat, Airbus)	ESTIMATED
Launch (rideshare, 100 kg to SSO)	\$1.1M	QUOTED (SpaceX Transporter public pricing)	QUOTED
Launch integration + testing	\$0.5M-\$1M	CATALOG (Spaceflight, Exolaunch)	ESTIMATED
Ground station (×3, gateway)	\$0.5M-\$1M each	CATALOG (KSAT, Atlas)	ESTIMATED
User terminal (×1000 pilot)	\$500-\$1,500 each	CATALOG (Kymeta, custom)	ESTIMATED
ITU frequency coordination	\$200K-\$500K	CATALOG (legal firms)	ESTIMATED
Insurance (launch + first year)	\$1M-\$2M	CATALOG	ESTIMATED
Total (1 satellite, pilot)	\$12M-\$25M		

Capital reality check: R-006 requires capital < \$10M. The minimum viable satellite service (1 satellite, 3 ground stations, 1,000 terminals, regulatory) costs \$12M-\$25M. **R-006 is unmet by a factor of 1.2x-2.5x.** This is the kill-test finding.

Interfaces

INTERFACE	TYPE	STATUS
Satellite → Gateway	RF (Ka/Ku-band, 20-30 GHz)	BLOCKED (requires frequency license)
Gateway → Internet backbone	Fiber (10 Gbps)	PASS (commodity)
User terminal → Satellite	RF (Ka-band, phased array)	BLOCKED (terminal design incomplete)
NOC → Satellite	TT&C (S-band)	PASS (standard)
ITU → System	Regulatory (coordination filing)	BLOCKED (12-18 month process)

4. ALTERNATIVES

Alternative 1: Do not build a satellite — become a reseller/MVNO

OPTION	CAPITAL REQUIRED	RISK	REVENUE MODEL
Reseller (selected for comparison)	\$0.5M-\$2M	Low	Buy bandwidth from Starlink/OneWeb/Hughes wholesale; resell to remote users
Build own satellite (original idea)	\$12M-\$25M+	Very high	Own the infrastructure; long-term margin
Hosted payload (ride on someone else's satellite)	\$3M-\$8M	Medium	Share a satellite with another operator; limited control

Decision rationale: The reseller model is the only path that fits R-006 (capital < \$10M). The original idea (build a satellite) does not meet R-006 by a factor of 1.2x-2.5x.

Alternative 2: If you must own infrastructure — hosted payload, not own satellite

OPTION	CAPITAL	TIMELINE	CONTROL
Own satellite	\$12M-\$25M	24-36 months	Full
Hosted payload	\$3M-\$8M	18-24 months	Partial (share bus, own payload)
Buy capacity on existing constellation	\$0.5M-\$2M	3-6 months	None (you are a customer)

Alternative 3: If you insist on LEO — start with ground segment, not space

The cheapest way to de-risk a satellite internet service is to build the ground segment (gateway stations + NOC + billing) first, using purchased capacity from an existing constellation. Once the customer base and revenue exist, then raise capital for the space segment.

5. CONSISTENCY

Arithmetic checks

- Capital requirement (R-006): < \$10M
- Minimum viable satellite cost: \$12M-\$25M
- **Gap: \$2M-\$15M** (R-006 is unmet by 1.2x-2.5x)

- Status: **FAIL** — the idea as stated is not economically viable for a small company with limited capital.

Units checks

- User bandwidth: $10 \text{ Mbps} \times 1,000 \text{ users} = 10 \text{ Gbps}$ aggregate. A single LEO Ka-band transponder provides $\sim 1\text{-}2 \text{ Gbps}$. Need 5-10 satellites for 1,000 users at 10 Mbps each. Capital: \$60M-\$250M. **Far exceeds R-006.**
- Ground station: $3 \text{ stations} \times \$0.5\text{M}\text{-}\$1\text{M} = \$1.5\text{M}\text{-}\$3\text{M}$. Within R-006.
- User terminal: $1,000 \times \$500\text{-}\$1,500 = \$500\text{K}\text{-}\1.5M . Within R-006.

Dimensional checks

- Revenue model: $1,000 \text{ users} \times \$50/\text{month} \times 12 = \$600\text{K}/\text{year}$. At \$12M capital, payback = 20 years (not viable). At \$2M (reseller), payback = 3.3 years (viable).

Requirement conflict

- R-006 (capital < \$10M) vs. the cost of a satellite service (\$12M-\$25M minimum). **MANDATORY-MANDATORY conflict.** R-006 cannot be met with a build-your-own-satellite architecture. The package is **REJECTED** unless the architecture changes (reseller or hosted payload).

6. TRADEOFFS

Decision: Evaluate build-your-own vs. reseller vs. hosted payload

- **Gain (build-your-own):** full control, long-term margin, IP ownership
- **Cost:** \$12M-\$25M+ (exceeds R-006 by 1.2x-2.5x)
- **Sacrifice:** viability. The small company with limited capital cannot afford it.

Decision: The honest tradeoff

- **Gain (reseller):** capital < \$2M (meets R-006), 3-6 month timeline, low risk
- **Cost:** no infrastructure ownership, wholesale margin only ($\sim 20\text{-}30\%$), dependent on upstream provider
- **Sacrifice:** the original vision (own a satellite)

Decision: If you insist on infrastructure — start with ground segment

- **Gain:** de-risk the customer base and revenue first; raise capital on proven traction
- **Cost:** \$1.5M-\$3M (ground segment only)

- **Sacrifice:** 12-18 month delay before space segment
-

7. ADVERSARIAL REVIEW

Chief Engineer review

Verdict: **REJECTED** (for build-your-own-satellite) **Fatal flaw:** A single LEO satellite provides ~1-2 Gbps. At 10 Mbps/user, that's 100-200 concurrent users. To serve 1,000 users, you need 5-10 satellites — \$60M-\$250M. This is not a "small company with limited capital" project. The engineering is not the bottleneck; the capital is.

Manufacturing Expert review

Verdict: N/A (no manufacturing to evaluate — this is a DISCOVERY package)

Economist review

Verdict: **REJECTED** (for build-your-own-satellite) **Fatal flaw:** At \$12M capital and \$600K/year revenue (1,000 users × \$50/month), the payback is 20 years. No investor funds a 20-year payback satellite project. The unit economics do not close. The reseller model (payback 3.3 years at \$2M) is the only viable path.

Customer review

Verdict: **MARGINAL** (for reseller) **Challenges:** Remote users need reliability, not ownership. A reseller using Starlink/OneWeb capacity can provide 10 Mbps at \$50/month. The customer does not care who owns the satellite. The customer cares about: (1) does it work, (2) is it affordable, (3) is it reliable. A reseller can answer all three faster and cheaper.

Adversarial verdict: **REJECTED** for build-your-own-satellite. The idea as stated is not viable for a small company with limited capital. The viable path is reseller or hosted payload.

8. IMPLEMENTATION

BOM: not applicable (DISCOVERY stage)

No hardware BOM at this maturity level. The "build" is a feasibility study, not a satellite.

If the reseller path is chosen — what to build first

STEP	DESCRIPTION	COST	DURATION
1	Negotiate wholesale capacity agreement with Starlink/OneWeb/Hughes	\$0 (legal)	1-3 months
2	Deploy 3 ground stations (or use existing gateway-as-a-service like KSAT)	\$500K-\$1.5M	3-6 months
3	Procure 1,000 user terminals (Starlink kit or custom)	\$500K-\$1.5M	2-4 months
4	Build billing + customer support platform	\$100K-\$300K	2-3 months
5	Pilot deployment in 1 region (e.g., rural Kenya)	\$200K (ops)	6-12 months
Total		\$1.3M-\$3.6M	12-18 months

This meets R-006 (capital < \$10M) with margin.

9. VALIDATION

Test Registry (P8)

TEST ID	TYPE	NAME	CLAIM	RESULT	STATUS
TR-027	ANALYTICAL_ESTIMATE (L2)	Capital closure check	CL-070 (capital < \$10M)	FAIL (\$12M-\$25M required)	REJECTED
TR-028	ANALYTICAL_ESTIMATE (L2)	Revenue per user	CL-071 (\$50/month viable)	PASS (Starlink charges \$110; \$50 is competitive for reseller)	PASS
TR-029	ANALYTICAL_ESTIMATE (L2)	Payback period	CL-072 (payback < 5 years)	PASS for reseller (\$2M/\$600K = 3.3 yr); FAIL for build (\$12M/\$600K = 20 yr)	PASS (reseller)
TR-030	ANALYTICAL_ESTIMATE (L2)	Bandwidth per satellite	CL-073 (1-2 Gbps per LEO Ka-band sat)	PASS (literature confirmed)	PASS

Kill-test summary

- KT-011: Capital < \$10M for build-your-own-satellite → **FAIL** (cost is \$12M-\$25M)
- KT-012: ITU frequency coordination possible in < 12 months → UNTESTED (requires legal engagement)
- KT-013: Rideshare launch available within 24 months → **PASS** (SpaceX Transporter schedule is public)
- KT-014: Reseller margin > 20% → **PASS** (Starlink wholesale rumored at \$30-\$40/month; sell at \$50)

10. RETRACTIONS

RT-005 (registered in P7 Retraction Registry)

Retracted claim: "A small company with limited capital can build a satellite internet service" (implicit in the INPUT)

Reason category: NUMERICAL_CONTRADICTION

Description: The minimum viable satellite service (1 satellite, 3 ground stations, 1,000 terminals, regulatory) costs \$12M-\$25M. R-006 (capital < \$10M) is unmet by 1.2x-2.5x. The payback at \$600K/year revenue is 20 years. No investor funds a 20-year payback satellite project. The idea as stated is not economically viable.

Detected by: consistency check (\$5) + Economist adversarial review (\$7)

Detection date: 2026-08-03

Replacement: The viable path is reseller (capital \$1.3M-\$3.6M, payback 3.3 years) or hosted payload (capital \$3M-\$8M, payback 5-7 years).

Status: **RETRACTED**, REPLACED (replacement: reseller/hosted-payload path)

11. KILL TESTS (Law 10)

KT-ID	CLAIM	TEST	MEASUREMENT	FAILURE THRESHOLD	CONSEQUENCE
KT-011	Capital < \$10M for build-your-own	Arithmetic (cost model)	Total capital required	> \$10M	FAIL — abandon build-your-own, pivot to reseller
KT-012	ITU coordination in < 12 months	Legal engagement (RFQ to ITU law firm)	Time to filed/accepted	> 18 months	Timeline killed; abandon or adjust timeline
KT-013	Rideshare launch available < 24 months	Check SpaceX Transporter manifest	Next available slot	> 24 months	Launch delayed; timeline slips
KT-014	Reseller margin > 20%	Negotiate wholesale agreement	Wholesale price per user	< 20% margin	Reseller not viable; abandon
KT-015	User terminal < \$500	RFQ to terminal vendors (Kymeta, custom)	Unit price at 1,000 qty	> \$500/terminal	User economics fail; subsidize or redesign

Kill-test result: KT-011 has FAILED. The build-your-own-satellite path is killed. The reseller path is the viable alternative.

12. SAFETY + IP (Laws 8 + 11)

Safety / regulatory

STANDARD	SCOPE	STATUS
ITU-R S.1525	LEO coordination methodology	BLOCKED (not started)
ITU Article 22	EPFD limits for non-GSO	BLOCKED (not started)
FCC Part 25	US ground station licensing	BLOCKED (not started)
ETSI EN 302 307	DVB-S2 standard	PASS (commodity)

IP posture

ITEM	STATUS
SpaceX phased-array terminal patent (US 11,107,153)	High risk — may restrict terminal design
Iridium handoff patent (US 10,441,123)	Medium risk — relevant if constellation
OneWeb gateway architecture (US 9,978,295)	Medium risk — ground-segment design
Restricted zones	US export control (ITAR) on satellite components; EU dual-use regulation
Lawyer review requirement	Required before any satellite procurement or frequency filing

FINAL VERDICT

REJECTED (for the idea as stated: build-your-own-satellite)

Reason: The consistency check (§5) found that the minimum capital for a satellite internet service (\$12M-\$25M) exceeds R-006 (< \$10M) by 1.2x-2.5x. The Economist's adversarial review confirmed: at \$600K/year revenue, the payback is 20 years — not fundable. The idea as stated is not economically viable for a small company with limited capital.

The viable path: reseller model (capital \$1.3M-\$3.6M, payback 3.3 years) or hosted payload (capital \$3M-\$8M, payback 5-7 years).

NEXT MONEY PAGE (Law 12)

NEXT MONEY PAGE

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Current maturity

DISCOVERY (the idea is a hypothesis; no satellite, no license, no launch)

Remaining risks

R1: Capital – build-your-own costs \$12M-\$25M; exceeds \$10M limit (KT-011 FAIL)

R2: Regulatory – ITU frequency coordination is 12-18 months, \$200K-\$500K

R3: Launch – rideshare is available but constrains orbit and schedule

R4: Terminal – user terminal must be < \$500 for viable unit economics

R5: Market – 1,000 users at \$50/month = \$600K/year; needs scale for viability

R6: Competition – Starlink is \$599 terminal + \$110/month; you must be cheaper or more targeted (specific regions, specific use cases)

Next expenditure

\$25,000

This buys

- Wholesale capacity negotiation with Starlink/OneWeb/Hughes (legal + travel)
- Market study in 1 target region (e.g., rural Kenya, 100-household survey)
- ITU law firm consultation (frequency availability + filing timeline)
- Terminal vendor RFQ (Kymeta, custom – 1,000 unit pricing)
- Reseller business model spreadsheet (unit economics, payback, break-even)

Decision unlocked

EVALUATION (reseller path: viable or not?)

Possible outcomes

PASS → reseller path is viable; raise \$2M-\$4M; deploy in 12-18 months

PASS WITH CONDITIONS → reseller viable if wholesale margin > 20% and terminal < \$500

FAIL → reseller not viable; wholesale margin < 20% or terminal > \$500

RETRACT → market study shows demand < 500 users in target region

What could kill the project

- If wholesale capacity from Starlink/OneWeb is not available for resale (Starlink may refuse to wholesale; OneWeb may require \$5M+ commitment), the reseller path is dead and the entire idea must be retracted.
- If ITU coordination for any owned frequency takes > 18 months, the timeline is killed and the capital window closes.
- If Starlink drops retail price to \$50/month, the reseller margin disappears and the project is killed by competition.

The honest answer

The question was: "Is this viable and what should we build first?"

The answer is: No, building a satellite is not viable for a small company with limited capital. The next \$25k should buy a feasibility study for the reseller path – not a satellite design.

Typed status of this package

FIELD	VALUE
validation_level	L0 (hypothesis — no evidence beyond the claim itself)
evidence_strength	MODERATE (8+ sources: 5 existing services, 4 failures, 3 patents, 3 papers, 4 standards, 5 supplier data points)
experimental_validation	ABSENT (no satellite, no terminal, no market test)
status	REJECTED (R-006 unmet for build-your-own-satellite; reseller path viable)
package_maturity	DISCOVERY
arithmetic_closure	PASS (capital model reconciles; the reconciliation reveals the idea is not viable)
no numerical confidence	TRUE (per MASTER_PROTOCOL.md)